Supporting Information

GitHub Repo

Processed result shows 7,286 total rows, 481 normal rows, and 6,805 faulty rows, so the dataset is highly imbalanced toward faults. The best unsupervised separation in the archived subset came from t-SNE + HDBSCAN and UMAP-fallback + HDBSCAN, while the supervised models all performed similarly, with about 93.3–93.4% accuracy and macro-F1 near 0.966.

EDA figures

A.1 Overview

This appendix explains the visual outputs generated from the archived Tennessee Eastman Process (TEP) study conducted in the context of chemical process automation for acetic acid production via oxidation of acetaldehyde. TEP00 represents the normal operating benchmark, while TEP01–TEP15 represent faulted operating conditions. Analysis was harmonized to a common 50-sensor feature space. The purpose of the figures is to show how exploratory analysis, dimensionality reduction, clustering, self-organizing maps, and supervised learning reveal differences between normal and abnormal process behavior.

A.2 Exploratory Data Analysis Figures

Figure A1. eda_class_balance.png

This figure shows the number of observations associated with each operating condition. Its main purpose is to demonstrate that the archived dataset is class-imbalanced, with fault cases collectively contributing many more observations than the single normal benchmark. This is important because class imbalance can influence clustering quality, classifier bias, and interpretation of predictive performance.

Figure A2. eda_rows_per_file.png

This figure compares the number of rows in each TEP file. It confirms that most trajectories are of similar length, while some files differ slightly. This matters because dynamic methods such as LSTM, DPCA, and MPCA are sensitive to sequence length and time structure.

Figure A3. eda_missing_fraction.png

This figure summarizes the proportion of missing values by file or sensor. Its main role is to show that the archived dataset was sufficiently complete for analysis and that the preprocessing stage did not rely heavily on imputation. In process-monitoring studies, this supports the validity of downstream machine learning results.

Figure A4. eda_sensor_variance.png

This plot ranks process variables by variance. Sensors with larger variance show stronger fluctuations across operating conditions, making them more likely to contribute to fault separation. In practical terms, this figure helps identify variables whose responses are most dynamic under plant disturbances.

Figure A5. eda_correlation_heatmap.png

This figure displays the correlation matrix among sensors. It highlights the strong interdependence of process variables, which is typical in industrial chemical systems with recycle streams, coupled control loops, and interacting unit operations. This supports the use of multivariate analysis rather than univariate monitoring.

Figure A6. eda_fault_pattern_heatmap.png

This heatmap shows how standardized sensor values change across the normal case and the different fault conditions. It provides a compact visual signature of each fault and helps identify whether different faults produce similar or distinct process responses.

Figure A7. eda_normal_faulty_sensor_delta.png

This figure compares average sensor behavior between the normal benchmark and the combined faulty group. It is one of the clearest visuals for identifying which variables shift the most under abnormal operation, making it especially useful for engineering interpretation.

Figure A8. sensitivity_top20.png

This figure summarizes the most influential variables from the sensitivity analysis. It helps identify which measurements most strongly distinguish normal from abnormal operation and therefore suggests which sensors may be most useful for monitoring and diagnosis.

A.3 Dimensionality Reduction and Embedding Figures

Figure A9. pca_subset_embedding.png

This figure shows the PCA projection of the archived subset. PCA is a linear method, so its role is to reveal whether the dominant directions of variance also correspond to fault separation. The figure indicates some structure, but not the clearest separation among all methods.

Figure A10. tsne_subset_embedding.png

This figure shows the t-SNE embedding. It is especially useful for visualizing local neighborhoods and often reveals compact groupings that PCA cannot. In the archived results, t-SNE produced one of the clearest visual separations among fault patterns.

Figure A11. umap_fallback_subset_embedding.png

This figure represents the archived UMAP-style fallback embedding. Its role is similar to UMAP: preserve local structure while still reflecting broader geometric relationships. It helps determine whether faults form interpretable manifolds in reduced space.

Figure A12. pacmap_fallback_subset_embedding.png

This figure shows the archived PaCMAP-style fallback representation. It is useful for comparing whether mid-range relationships among samples reveal better group structure than PCA or t-SNE alone.

Figure A13. dpca_subset_embedding.png

This figure shows Dynamic PCA, which incorporates time-lag information. Its importance lies in the fact that faults in process plants evolve over time, so DPCA can reveal temporal structure that ordinary PCA may miss.

Figure A14. trimap_fallback_subset_embedding.png

This figure shows the TriMAP-style fallback representation. It is intended to preserve broader neighborhood relationships and therefore offers another perspective on global fault geometry.

Figure A15. mpca_subset_embedding.png

This figure presents the MPCA embedding. MPCA is useful when variability has grouped or structured components, and the figure helps assess whether such grouped behavior improves fault separation.

Figure A16. autoencoder_subset_embedding.png

This figure shows the learned latent representation from an autoencoder. It evaluates whether nonlinear neural compression can organize the high-dimensional sensor space into a more fault-separable form.

Figure A17. autoencoder_subset_loss.png

This plot shows the training loss of the autoencoder. A decreasing loss indicates that the model successfully learned a compact representation capable of reconstructing the input data with increasing accuracy.

A.4 Clustering Figures

Figures A18–A25. K-means clustering figures

These figures show K-means clustering applied to each reduced representation. Their purpose is to test whether centroid-based partitions can separate normal and faulty regions once the data are embedded into a lower-dimensional space. Better separation suggests that the embedding produces compact, approximately spherical operating-state groupings.

Figures A26–A33. HDBSCAN clustering figures

These figures show HDBSCAN clustering on the different embeddings. HDBSCAN is valuable because it identifies dense regions without forcing every point into a cluster. In the archived

results, HDBSCAN performed especially well on nonlinear embeddings such as t-SNE and the UMAP-style fallback.

Figures A34–A42. Hierarchical clustering figures

These figures show hierarchical clustering results and, where provided, dendrogram structure. Their main contribution is interpretability: they reveal nested relationships among operating conditions and show which faults appear more similar to one another before separating into distinct groups.

A.5 SOM Figures

Figure A43. som_subset_u_matrix.png

The SOM U-matrix visualizes distances between neighboring neurons. Large distances indicate boundaries between distinct operating regions, so this figure helps identify transitions between normal and fault-dominated areas of the process map.

Figure A44. som_subset_hit_map.png

This figure shows how frequently different SOM neurons are activated by the data. Dense hit regions indicate common process states, while sparse or isolated regions suggest unusual or rare fault behaviors.

A.6 Supervised Learning Figures

Figure A45. rf_confusion_matrix.png

This confusion matrix summarizes Random Forest predictions. Most observations lie on the diagonal, showing strong classification performance in distinguishing normal from faulty operation.

Figure A46. gb_confusion_matrix.png

This figure shows the Gradient Boosting confusion matrix. Its pattern is very similar to the Random Forest result, indicating comparable predictive strength.

Figure A47. lstm_confusion_matrix.png

This confusion matrix summarizes LSTM predictions using temporal sequences. It demonstrates that dynamic modeling can also separate normal and abnormal operation effectively, though in the archived study it did not dramatically exceed the tree-based models.

Figure A48. lstm_training_loss.png

This figure shows the LSTM training-loss trajectory. A stable downward trend suggests that the model learned useful temporal process features without severe instability.

Figure A49. supervised_model_comparison.png

This figure compares the supervised models directly. The main conclusion is that Random Forest, Gradient Boosting, and LSTM all achieved strong performance, with the tree-based models slightly outperforming the LSTM in this archived run.

A.7 **Overall Interpretation**

Taken together, the figures show that the archived TEP00–TEP15 study successfully identified meaningful process structure in both unsupervised and supervised settings. Nonlinear dimensionality reduction combined with density-based clustering revealed clearer fault organization than purely linear methods, while supervised models achieved strong normal-versus-fault classification. From an industrial perspective, the figures support the idea that multivariate sensor data can be used to detect, visualize, and interpret abnormal operating behavior in complex chemical process systems.

Bottom-line interpretation

For the TEP00–TEP15, the figures collectively say this:

- the fault data are strongly dominant over the normal case,
- only a subset of sensors appears to drive most separability,
- nonlinear embeddings reveal clearer fault structure than PCA alone,
- t-SNE + HDBSCAN and UMAP-style + HDBSCAN gave the strongest unsupervised grouping in the archived metrics,
- supervised binary normal/fault detection was strong across all three models, at roughly 93.3–93.4% accuracy,
- and the overall workflow supports the industrial interpretation that abnormal operating modes in a coupled chemical process can be identified from multivariate sensor behavior.